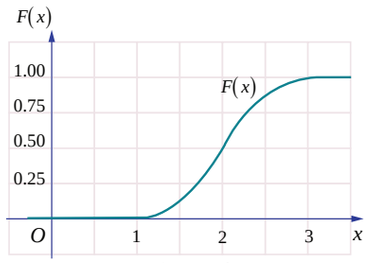
$$= 0 + \left[\frac{(u-1)^2}{2}\right]_1^2 + \left[-\frac{(3-u)^2}{2}\right]_2^3 + 0$$

$$= \frac{1}{2} + \frac{1}{2} = 1.$$

These give $F(x) = \begin{cases} 0, & -\infty < x < 1 \\ \dfrac{(x-1)^2}{2}, & 1 \le x < 2 \\ 1 - \dfrac{(3-x)^2}{2}, & 2 \le x < 3 \\ 1 & 3 \le x < \infty \end{cases}$



Distribution function

Fig. 11.17

(ii) $P(1.5 \le X \le 2.5) = F(2.5) - F(1.5)$

$$= \left(1 - \frac{(3-2.5)^2}{2}\right) - \left(\frac{(1.5-1)^2}{2}\right)$$

$$= \frac{1.75 - 0.25}{2} = 0.75$$

or

$$P(1.5 \le X \le 2.5) = \int_{1.5}^{2.5} f(x)\,dx = \int_{1.5}^{2} (x-1)\,dx + \int_{2}^{2.5} (-x+3)\,dx = 0.75.$$

Check: (i) Whether $F(x)$ is continuous everywhere.

(ii) From the Fig. 11.16, triangle area $= \frac{1}{2}bh = 1$.

11.4.5 Probability density function from Probability distribution function.

Let us learn the method to determine the probability density function $f(x)$ from the distribution function $F(x)$ of a continuous random variable X.

Suppose $F(x)$ is the distribution function of a continuous random variable X. Then the probability density function $f(x)$ is given by

$$f(x) = \frac{dF(x)}{dx} = F'(x), \text{ wherever derivative exists.}$$

Example 11.13

If X is the random variable with distribution function $F(x)$ given by,

$$F(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \le x < 1 \\ 1, & 1 \le x \end{cases}$$

then find (i) the probability density function $f(x)$ (ii) $P(0.2 \le X \le 0.7)$.

12th_Maths_Vol 2_EM_CH 11_Probability Distributions_29-05-20.indd 199 16-12-2022 16:51:19